OncoAI

OncoAI: The AI for Oncologists

Presented by Roy Aggarwal

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- Instructor: Reeta Suman
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Application Overview

- The (fictional) Onco AI-powered app is a human centred medical scribe solution
- OncoAI interprets clinical conversations and generates structured medical documentation
- AI-generated reports save oncologists hours of non-medical administrative work

The Healthcare Decision-Making Challenge

- The key challenge is to gain clinician trust by being consistently accurate and reliable
(van Harten et al., 2022; World Health Organization, 2021)

- This matters because trust is key to adoption and successful implementation

- Real World Impact: Without clinician trust, the application would be unused and thus worthless

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Problem Scope & Relevance

- Hallucination is a major safety concern in AI-generated clinical notes (Johannessen et al., 2025)
- Due to extensive model training, OncoAI has an industry leading hallucination rate of only 0.05%. This equates to one in 2000 reports affected on average.
- Contrast this to hallucination rates of 7-25% for similar products in the market

Model Choice: Which is the most suitable?

The shortlist

- Clinical NLP / Information Extraction Models
- Automatic Speech Recognition (ASR) Models
- Large Language Models (LLMs)

The Winner: LLM

LLMs are best at summarizing conversations into structured clinical documentation

Human-Centred AI Principles

- User Needs: Oncologists need reliable and accurate documentation throughout the treatment lifecycle
- Human Oversight: No matter how good any AI is, there will always be a need for human validation
- Transparency: Doctors can clearly see, understand, and verify how the AI produced its output

(World Health Organization, 2021)

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Data Collection

- Types of data needed: Patient pathology data
- Data sources: Historical lab results, imaging results, patient interview notes, peer reviews
- Diversity and representativeness: Ability to differentiate different cancer types & stages (Zhang et al., 2026)

AI Model Design

- Model Type: LLM trained on a very large set of oncology-specific data (Zhang et al., 2026).
- The model supports decision making by providing a "second pair of eyes" as input into treatment decisions
- Continuous Improvement: The model is refined almost constantly since new oncology field research emerges almost constantly

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Human-Centred Design Integration

- User interviews: Regular engagement with oncologists to keep the tool clinically relevant.
- Prototyping: Iterative vendor–clinician cycles to refine workflows and note structure.
- Usability testing: Tightly integrated with prototyping as part of iterative software development methodologies (World Health Organization, 2021).

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Ethical Considerations

- Bias & fairness: Avoid misinterpreting symptoms across diverse patient groups

- Explainability: Clinicians can see what the AI heard and how it summarized it.

- Accountability: Final documentation always reviewed and approved by the oncologist (World Health Organization, 2021).

•Data protection: All recordings and notes encrypted end-to-end

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Privacy, Security & Consent

- Data Protection: Encrypted recordings and notes
- Consent Workflow: Explicit patient consent before any visit is captured, and possibly every visit
- Follow provincial & federal healthcare privacy laws for sensitive medical patient data, as a minimum standard (HealthOrbit AI, 2025).

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Understanding End-Users

- Clinician needs: Reduce documentation load, increase documentation accuracy and thoroughness (van Harten et al., 2022).

- Patient needs: Clear, respectful and complete capture of symptoms and concerns during visits

- Workflow alignment: The app fits naturally into oncology consultations without slowing care

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User Interface Design

- UI principles: Simple & intuitive layout with minimal clicks during consults. Always keep the user in mind
- Accessibility: Clear text, high contrast, suitably large fonts, optional speech-to-text capabilities for dictation
- Example elements: Quick-edit panels for research summaries, symptom lists, and treatment plans

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Diverse User Demographics

- Cultural factors: The app supports different communication styles from different oncologists
- Multi Language Support: The app handles accents and multilingual doctor/patient interactions
- Accessibility: Works for clinicians with different tech comfort levels, adapts to a clinician’s style over time (World Health Organization, 2021).

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Model Validation & Testing

- Accuracy testing: Compare AI-generated notes to clinician-verified oncology notes, track hallucination/error rates over time (Johannessen et al., 2025).
- Real-world validation: Test during live consults across different cancer types. Report errors to manufacturer. Especially important for early adopters
- Limitations: May miss nuances in medically complex or emotionally sensitive discussions

Limitations & Uncertainties

- Data gaps: May miss rare cancer terminology or atypical symptom descriptions
- Model uncertainty: Struggles with ambiguous or emotionally complex dialogue
- Human–AI limits: Requires ongoing clinician review to catch subtle clinical nuances (Johannessen et al., 2025).

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Lessons Learned & Future Improvements

- Key insights: AI reduces oncology note burden but still needs careful clinician oversight (van Harten et al., 2022).
- Next steps: Improve handling of rare cancers and emotionally complex discussions

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AI Usage Statement

- I used AI to create a skeleton with bullet points that match the requirements of the assignment & rubric. I also used AI occasionally to flesh out an idea. But the content is mine with occasional suggestions from AI.

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Reference List

HealthOrbit AI. (2025, February 9). *Are your medical records safe with AI scribes?* [Blog post]. HealthOrbit AI.

Johannessen, J. A., et al. (2025). A framework to assess clinical safety and hallucination rates of large language models for clinical note generation. *NPJ Digital Medicine*, 8(1), 45.

van Harten, W. H., et al. (2022). Quantifying the electronic health record burden in head and neck cancer care. *Head & Neck*, 44(11), 2533–2542.

World Health Organization. (2021). *Ethics and governance of artificial intelligence for health*. World Health Organization.

Zhang, X., et al. (2026). AI and big data in oncology: A physician-centered perspective on multimodal data integration. *The Oncologist*, 31(2), 123–137.

Stults, C. D., et al. (2025). An ambient artificial intelligence documentation platform for clinicians. *JAMA Network Open*, 8(4), e2312345.

Patel, N., Chen, Y., & Rossi, L. (2025). Enhancing clinical documentation with ambient artificial intelligence: A survey of clinician experiences. *JAMIA Open*, 8(1), ooaf013.

Lee, S., Kumar, A., & Martinez, J. (2025). Large language models in medicine: Clinical applications, technical challenges, and ethical considerations. *NPJ Digital Medicine*, 8(1), 76.

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Questions

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